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Time to Practice: Singular Value Decomposition (SVD)
Graded Assignment • 15 min

Due Oct 14, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the primary outcome of applying Singular Value Decomposition (SVD) to a centered data matrix? 1 / 1 point

- Linear Regression Analysis
- Cluster Analysis
- Principal Components Analysis (PCA)

Nice work

When a data matrix is centered and Singular Value Decomposition is applied, the result is Principal Components Analysis. This is because SVD extracts the principal components from the centered data matrix.

2. How does truncated Singular Value Decomposition help in data compression? 1 / 1 point

- By increasing data storage
- By reducing dimensionality
- By enhancing data resolution
- By encrypting the data

Nice work

Truncated SVD helps in data compression by reducing the dimensionality of the data. It approximates the original matrix with a matrix of lower rank, thus capturing the most significant data features while using fewer data.

3. Why is Singular Value Decomposition considered an effective tool for matrix approximation? 1 / 1 point

- It is specialized to handle input matrices of low rank
- It minimizes a least-squares error criterion
- It can exploit matrix symmetry

Nice work

Correct. Singular Value Decomposition provides the best possible approximation of the original matrix in the least squares sense, making it highly effective for data compression and reduction.